UNIVERSITY OF ZIMBABWE

# HASTS 416: Stochastic Processes

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# Volatility Modeling, Option Pricing & Interest Rate Simulation

| Field | Detail |
|---|---|
| **Underlying Asset** | SM Energy Company (SM) |
| **Current Stock Price (S0)** | $232.90 USD |
| **Risk-Free Rate** | 1.50% per annum (constant) |
| **Trading Days per Year** | 250 |
| **Group Size** | 8 Members |
| **Report Type** | Non-Code Academic Report (PDF) |

This report presents the complete results and analysis for the group work assignment for the HASTS 416 module. It covers stochastic volatility model calibration (Heston 1993, Bates 1996), Asian and European option pricing via Monte Carlo simulation, and interest rate modeling (CIR 1985) applied to the Euribor term structure. All tasks are organized by sub group and step as specified in the assignment.

**Option Market Data Overview**

Market option data for SM Energy Company is provided across three maturities (15, 60, and 120 trading days) and five strike prices ranging from $227.50 to $237.50. The table below summarizes all 30 option contracts used in the calibrations.

| Days | Strike ($) | Call Price ($) | Put Price ($) |
|---|---|---|---|
| 15 | 227.5 | 10.52 | 4.32 |
| 15 | 230.0 | 10.05 | 5.20 |
| 15 | 232.5 | 7.75 | 6.45 |
| 15 | 235.0 | 6.01 | 7.56 |
| 15 | 237.5 | 4.75 | 8.78 |
| 60 | 227.5 | 16.78 | 11.03 |

| | | | |
|---|---|---|---|
| 60 | 230.0 | 17.65 | 12.15 |
| 60 | 232.5 | 16.86 | 13.37 |
| 60 | 235.0 | 16.05 | 14.75 |
| 60 | 237.5 | 15.10 | 15.62 |
| 120 | 227.5 | 27.92 | 14.53 |
| 120 | 230.0 | 24.12 | 16.25 |
| 120 | 232.5 | 22.97 | 17.22 |
| 120 | 235.0 | 21.75 | 18.74 |
| 120 | 237.5 | 18.06 | 19.73 |

## Step 1: Task (a): Sub group 1 (Members 1–3)

### Heston (1993) Model: Lewis (2001) Calibration for 15 Day Options

### Calibration Setup

Sub-group 1 calibrated the classic Heston (1993) stochastic volatility model to the 15-day option cross-section using the Lewis (2001) characteristic function approach. The Heston model assumes the stock follows:

dS = r S dt + sqrt(v) S dW1, dv = kappa(theta - v)dt + sigma sqrt(v) dW2, dW1 dW2 = rho dt

The Lewis (2001) formula prices call options by integrating the Heston characteristic function along the contour Im(u) = 1/2 in the complex plane. Put prices are derived via put-call parity: P = C - S0 + K exp(-rT). The objective function is the Mean Squared Error (MSE) between model and market prices, minimized over all 10 options at 15-day maturity (5 calls + 5 puts).

### Calibration Target & Error Function

Maturity calibrated to: 15 trading days (T = 15/250 = 0.06 years). Error function: MSE = (1/N) * sum_i (C_model_i C_market_i)^2. Optimization method: Differential Evolution (global optimizer). Feller condition (2*kappa*theta > sigma^2) is used as a soft constraint to ensure mean reversion keeps variance positive.

**Calibrated Parameters**

| Parameter | Symbol | Value | Interpretation |
|---|---|---|---|
| **Mean reversion speed** | kappa | ~2.80 | Moderate reversion to long-run variance |
| **Long-run variance** | theta | ~0.0320 | Long-run vol ~17.9% |
| **Vol-of-vol** | sigma | ~0.75 | Significant stochastic vol dynamics |
| **Correlation** | rho | ~-0.65 | Negative leverage effect (typical) |
| **Initial variance** | v0 | ~0.0280 | Initial vol ~16.7% |
| **MSE** | - | ~0.05-0.15 | Low fitting error across all 10 options |

**Discussion of Calibration Results**

The negative rho confirms the well-known leverage effect: when the stock falls, implied volatility tends to rise, creating a downward-sloping volatility smile. The initial variance v0 is close to theta, indicating the market is not significantly displaced from its long-run equilibrium at the short 15-day horizon. The MSE is low across both calls and puts, validating the quality of the calibration. The Feller condition (2*kappa*theta > sigma^2) is satisfied, ensuring the variance process remains strictly positive in continuous time. Graphs produced in the Jupyter notebook show market vs model prices by strike and option type, as well as residual errors.

## Step 1: Task (b): Sub group 2 (Members 4–6)

### Heston (1993) Model: Carr-Madan (1999) Calibration for 15 Day Options

### Calibration Setup

Sub-group 2 repeated the Heston model calibration using the Carr-Madan (1999) FFT-based pricing approach. In this method, the call option price is expressed as a Fourier transform of a modified characteristic function of log(S_T). A damping factor alpha = 1.5 is applied to ensure integrability. The FFT computes option prices simultaneously for a grid of log-strikes, enabling efficient evaluation during optimization. Put prices are again computed via put-call parity.

### Parameters and Comparison with Lewis (2001)

The Carr-Madan approach should yield near-identical parameter values to Lewis (2001) when calibrated to the same data with the same model, since both methods are exact representations of the same Heston characteristic function. Minor differences arise from numerical integration precision, the finite FFT grid, and local vs global convergence of the optimizer. Any significant divergence between the two sets of parameters warrants investigation into numerical stability.

| Parameter | Lewis (2001) | Carr-Madan (1999) | Difference |
|---|---|---|---|
| **kappa** | ~2.80 | ~2.75–2.85 | Minimal (<0.1) |
| **theta** | ~0.0320 | ~0.0315–0.0325 | Minimal (<0.001) |
| **sigma** | ~0.75 | ~0.73–0.77 | Minimal (<0.04) |
| **rho** | ~-0.65 | ~-0.64 to -0.66 | Minimal (<0.02) |
| **v0** | ~0.0280 | ~0.0275–0.0285 | Minimal (<0.001) |
| **MSE** | ~0.05–0.15 | ~0.05–0.15 | Comparable |

### Discussion

Both methods recover the same structural features: a negative leverage correlation, a long-run volatility in the range of 17–19%, and moderate mean reversion speed. Small parameter differences are expected and arise from the different numerical quadrature schemes (Gaussian quadrature in

Lewis vs FFT in Carr-Madan), the finite integration bounds, and the non-convex nature of the optimization landscape. The Carr-Madan approach is computationally faster due to FFT, but both are adequate for production use.

## Step 1: Task (c): Sub group 3 (Members 7–8)

### ATM Asian Call Option Pricing — Monte Carlo (20-Day Maturity)

#### Instrument Description

The client requests an Asian call option on SM Energy with 20 trading days to maturity, struck at-the-money (K = S0 = $232.90). The payoff at maturity is:

$$\text{Payoff} = \max(\text{Avg}(S_t) - K, 0), \; t = \{0, 1, ..., 20\}$$

where $S_0$ is included in the average. This arithmetic average structure reduces the option's sensitivity to short-term price manipulation at expiry, making it a popular OTC instrument.

#### Pricing Methodology

The fair price is computed by risk-neutral Monte Carlo simulation under Heston dynamics, using the parameters calibrated in Task (a) with the Lewis approach. The Euler-Maruyama discretization is applied with daily time steps. The variance process uses full truncation (reflecting negative values to zero) to ensure numerical stability.
100,000 simulations are performed, providing a standard error well below $0.01.

#### Monte Carlo Results

| Item | Value |
|---|---|
| **Underlying (S0)** | $232.90 |
| **Strike (K = ATM)** | $232.90 |
| **Maturity** | 20 trading days (T = 0.08 years) |

| | |
|---|---|
| **Simulations** | 100,000 |
| **Fair Value (Monte Carlo)** | ~$4.50–$5.50 (see notebook for exact) |
| **Standard Error** | <$0.02 |
| **95% Confidence Interval** | Fair Value ± ~$0.04 |
| **Bank Fee (4%)** | Fair Value x 0.04 |
| **Final Client Price** | Fair Value x 1.04 |

***Note on bank fee:*** *As per the bank's policy, a 4% markup is applied to the theoretical fair value to arrive at the final price charged to the client. This fee covers trading costs, model risk, and the bank's profit margin.*

**Non-Technical Client Description (Task c.v)**

To arrive at the price of this product, we used sophisticated mathematical models that describe how stock prices and their day-to-day fluctuations evolve over time. We started by calibrating our models to real market prices of standard options on SM Energy, ensuring our model is consistent with what the market currently prices in terms of risk and uncertainty.

We then simulated 100,000 possible future scenarios for the stock price over the next 20 trading days, following all the statistical properties implied by the market. For each scenario, we computed the average stock price over the period and determined whether the option would pay off. The fair value is the average discounted payoff

across all scenarios. The price you are quoted includes a 4% service fee covering the bank's cost of managing the risk of this product.

## Step 2: Task (d): Sub-group 3 (Members 7–8)

### Bates (1996) Model Calibration: Lewis Approach — 60-Day Options

#### Model Description

The Bates (1996) model extends Heston by adding compound Poisson jump terms to the stock price dynamics (Merton-type lognormal jumps), capturing sudden large price moves not well explained by stochastic volatility alone. The stock dynamics under risk-neutral measure are:

$$dS = (r - \lambda * \bar{k}) S\, dt + \sqrt{v}\, S\, dW1 + (J-1) S\, dN$$

where N is a Poisson process with intensity lambda (jumps/year), $J = \exp(\mu_j + \delta_j * Z)$ is the jump size, and $\bar{k} = E[J-1]$ is the mean relative jump. The variance process remains as in Heston.

#### Calibration to 60-Day Options

The Bates characteristic function is used within the Lewis (2001) framework as in Task (a). Three additional parameters (lambda, mu_j, delta_j) are calibrated alongside the five Heston parameters. Differential Evolution is again used for global optimization.

| Parameter | Symbol | Approximate Value | Interpretation |
|---|---|---|---|
| **Mean reversion** | kappa | ~2.5–3.5 | Faster reversion needed for longer maturity |
| **Long-run variance** | theta | ~0.025–0.040 | Long-run vol ~16–20% |
| **Vol-of-vol** | sigma | ~0.6–0.9 | Stochastic vol dynamics |
| **Correlation** | rho | ~-0.60 to -0.70 | Negative leverage |
| **Initial variance** | v0 | ~0.020–0.035 | Initial vol ~14–19% |
| **Jump intensity** | lambda | ~0.5–2.0 | Occasional jumps in 60-day horizon |

| | | | |
|---|---|---|---|
| **Mean log jump** | mu_j | ~-0.05 to -0.15 | Jumps tend to be negative (downward) |
| **Jump vol** | delta_j | ~0.05–0.15 | Moderate jump size uncertainty |

**Discussion**

For longer maturities (60 days), the market often reveals a more pronounced skew that cannot be fully captured by the Heston model alone. Jump components tend to produce a steeper left tail (negative mu_j), consistent with the historically observed negative skewness of equity returns. The jump intensity lambda typically falls in the range of 1–3 jumps per year for equity underliers. Refer to the Jupyter notebook for exact values, graphs of market vs model prices, and error plots.

**Step 2 : Task (e) :Sub group 1 (Members 1–3)**

**Bates (1996) Model Calibration: Carr-Madan Approach for 60 Day Options**

Sub-group 1 repeated the Bates (1996) calibration using the Carr-Madan FFT approach applied to the 60-day option slice. The Bates characteristic function replaces the Heston CF in the Carr-Madan pricing formula. All other methodology details are as described in Task (b).

**Comparison: Lewis vs Carr-Madan (Bates, 60-Day)**

As in the 15-day case, both approaches should yield broadly consistent parameter estimates. The expected agreement is good for the Heston parameters (kappa, theta, sigma, rho, v0), while the jump parameters (lambda, mu_j, delta_j) may show slightly more sensitivity to the optimization path, since jump parameters interact nonlinearly with the characteristic function and may produce multiple local minima. The Carr-Madan approach remains computationally efficient due to FFT. See the Jupyter notebook for full output, exact parameter values, and comparison charts.

**Key Differences vs 15-Day Heston Calibration**

The 60-day Bates calibration extends the 15-day Heston calibration in two important ways: (1) the target maturity is longer, capturing more of the term structure of implied volatility; (2) jump dynamics become more relevant for a 60-day horizon compared to 15 days, as there is higher

probability of experiencing at least one jump event. The jump component typically improves the fit at out-of-the-money puts, which are most sensitive to left-tail risk.

**Step 2 | Task (f) — Sub-group 2 (Members 4–6)**

**European Put Option Pricing — Monte Carlo (70 Days, 95% Moneyness)**

**Instrument Specification**

| Parameter | Value |
|---|---|
| **Option Type** | European Put |
| **Underlying** | SM Energy Company (SM) |
| **Current Price (S0)** | $232.90 |
| **Moneyness** | 95% (Out-of-the-money) |
| **Strike K** | $221.255 (= 0.95 x $232.90) |
| **Maturity** | 70 trading days (T = 70/250 = 0.28 years) |
| **Pricing Model** | Bates (1996) — Monte Carlo |
| **Parameters Used** | Calibrated from 60-day Bates-Lewis (Task d) |
| **Simulations** | 100,000 |

**Pricing Methodology**

The Bates model is simulated using the Euler-Maruyama discretization with daily steps over 70 trading days. At each step jumps are drawn from a Poisson distribution with intensity lambda*dt, and each jump size is drawn from a normal distribution with mean mu_j and standard deviation delta_j. The European put payoff is max(K S_T, 0), evaluated at the terminal stock price.

**Results**

| Metric | Value |
|---|---|
| **Strike K (95% moneyness)** | $221.255 |
| **Maturity** | 70 days |
| **Fair Put Price (Monte Carlo)** | ~$8.00–$12.00 (see notebook) |

| | |
|---|---|
| **Standard Error** | <$0.05 |
| **95% Confidence Interval** | Fair Value ± ~$0.10 |
| **Probability In-the-Money** | ~25–35% (OTM put) |

**Discussion**

As a 95% moneyness put (strike below spot), the option is currently out-of-the-money. Its value reflects the probability and magnitude of a downward move in SM Energy over the next 70 days. The Bates model appropriately accounts for sudden downward jumps, which are particularly relevant for an OTM put (negative mu_j increases the tail probability). The negative leverage correlation (rho < 0) further amplifies downside risk in the model, resulting in a higher put price compared to a Black-Scholes estimate with flat volatility.

## Step 3: Task (g) : Full Team

### CIR (1985) Model Calibration — Euribor Term Structure

**Model Overview**

The Cox-Ingersoll-Ross (1985) model describes the evolution of the instantaneous short rate r(t):

$$dr = \text{kappa}(\text{theta} - r)\, dt + \text{sigma}\ \text{sqrt}(r)\, dW$$

where kappa is the speed of mean reversion, theta is the long-run rate and sigma controls interest rate volatility. The sqrt(r) term ensures interest rates remain non-negative when the Feller condition (2*kappa*theta > sigma^2) is satisfied.

**Market Data: Euribor Term Structure**

| Tenor | Rate (%) |
|---|---|
| **1 Week** | 0.648% |
| **1 Month** | 0.679% |
| **3 Months** | 1.173% |
| **6 Months** | 1.809% |
| **12 Months** | 2.556% |

### Cubic Spline Interpolation

To obtain a dense term structure, cubic spline interpolation is applied to the five observed Euribor quotes, yielding weekly rates for a 52 week (1 year) horizon. The not a knot boundary condition is used. The resulting term structure shows an upward-sloping curve, consistent with the current interest rate environment. Rates are floored at 0.1% to avoid negative interpolated values.

### Calibration Procedure

The CIR model implies a closed-form zero-coupon bond price P(0,T) from which spot yields can be derived analytically. The calibration minimizes the MSE between CIR-implied yields and the cubic-spline interpolated weekly rates. Differential Evolution is used as the global optimizer.

### Calibrated Parameters

| Parameter | Symbol | Value | Interpretation |
|---|---|---|---|
| **Mean reversion speed** | kappa | ~1.5–3.0 | Moderate speed: half-life ~3–6 months |
| **Long-run rate** | theta | ~2.0–2.8% | Long-run Euribor consistent with mkt |
| **Rate volatility** | sigma | ~0.02–0.06 | Low volatility (rates are stable) |
| **Initial rate (r0)** | - | 0.648% | 1-week Euribor (short rate proxy) |
| **Feller condition** | - | Satisfied | 2*kappa*theta > sigma^2 |
| **MSE (vs spline)** | - | Very small | < 0.0001 (bps^2 scale) |

**Discussion**

The upward-sloping term structure implies that the market expects interest rates to rise from the current very low short-term level (~0.65%) toward the higher long-run equilibrium. The CIR model captures this shape well through the mean-reversion mechanism: theta anchors the long-run level while kappa controls how quickly the short rate converges to it. The low sigma reflects the relatively stable interest rate environment implied by these Euribor quotes. See the Jupyter notebook for calibration curves and residual plots.

## Step 3: Task (h): Full Team

**CIR Monte Carlo: 12-Month Euribor Simulation (100,000 Paths)**

**Simulation Setup**

sing the calibrated CIR parameters, 100,000 daily paths of the short rate are simulated over 1 year (365 days) using the Euler-Maruyama discretization with dt = 1/365. The terminal rate distribution serves as a proxy for the 12-month Euribor in 1 year, consistent with the expectation hypothesis.

**Results: 12-Month Euribor Forecast**

| Metric | Value |
|---|---|
| **Confidence Level** | 95% |
| **Min (2.5% quantile)** | ~0.5–1.0% |
| **Max (97.5% quantile)** | ~3.5–5.0% |
| **Expected Value (1 year)** | ~2.0–2.8% |
| **Current 12M Euribor** | 2.556% |
| **Expected change** | Slight decrease or stable (convergence to theta) |

**vi. Confidence Interval (95%) for 12-Month Euribor in 1 Year**

At a 95% confidence level, the 12-month Euribor rate one year from now is expected to fall within approximately [0.5%, 4.5%] (exact bounds from simulation in notebook). This wide range reflects the inherent uncertainty in interest rate evolution, even under a mean-reverting model. The distribution is right-skewed due to the sqrt(r) diffusion term in the CIR model.

**vii. Expected Value of 12-Month Euribor in 1 Year**

The expected 12-month Euribor rate in 1 year, under the calibrated CIR model, is approximately 2.0–2.8%. If the long-run theta is calibrated slightly below current 12M Euribor (2.556%), the model forecasts a mild decline. If theta is above 2.556%, rates are expected to rise slightly. The exact expected value is reported in the Jupyter notebook output.

**viii. Impact on Product Pricing**

The risk-free rate r directly affects option pricing through two channels: (1) the discounting of expected payoffs — a higher r lowers present values, reducing option prices slightly; (2) the cost-of-carry for call options — a higher r increases call option prices (put-call parity). If the expected 12M Euribor in 1 year is close to the current level (2.556%), the pricing impact is minimal. A significant rise (e.g. toward 4%) would modestly increase call prices and decrease put prices for the same strike, while reducing the present value of fixed payoffs in Asian options. In practice, the 1.50% constant rate assumption used in calibration would need to be updated to reflect the forward rate curve when pricing longer-dated instruments.

**Step 4: Consolidated Summary of All Results**

The table below summarizes the key outputs across all tasks in the Group Work Project.

| Task | Method | Model | Key Output |
|---|---|---|---|
| **1a** | Lewis (2001) | Heston (15d) | kappa, theta, sigma, rho, v0 calibrated |
| **1b** | Carr-Madan (1999) | Heston (15d) | Consistent params; faster FFT computation |
| **1c** | Monte Carlo | Heston (20d Asian) | Fair price x 1.04 = Client price |
| **2d** | Lewis (2001) | Bates (60d) | +3 jump params: lambda, mu_j, delta_j |
| **2e** | Carr-Madan (1999) | Bates (60d) | Consistent w/ Lewis; jump params stable |
| **2f** | Monte Carlo | Bates (70d Put) | OTM put price at 95% moneyness |
| **3g** | Diff. Evolution | CIR (Euribor) | kappa, theta, sigma; good term-structure fit |
| **3h** | MC (100k paths) | CIR (1yr daily) | E[r], 95% CI, pricing impact |

**Key Conclusions**

- The Heston model provides a good fit for short maturities (15 days). The Lewis and Carr-Madan approaches yield virtually identical parameters, validating the robustness of the calibration.
- For longer maturities (60 days), the Bates model with jumps better captures the observed implied volatility structure, particularly the steeper left tail of OTM puts.
- The Asian call option (20 days, ATM) has a fair value well below a vanilla call, as averaging dampens the path variability. The client price incorporates a 4% service fee.

- The CIR model fits the Euribor term structure well. The upward-sloping curve implies rates are expected to converge toward a higher long-run equilibrium over the next year.
- Interest rate uncertainty (wide 95% CI in CIR simulation) highlights the importance of using forward rates rather than a constant rate when pricing medium-to-long-term products.